Beyond Traditional Revenue Management: A Hybrid Analytics Framework for Strategic Growth Optimization in Tourism Startups

Martina Luzzi*[1], Carlo Giglio[1], and Francesca Guerriero[1]

[1]Department of Mechanical, Energy and Management Engineering, Unuversity of Calabria ITALY

*Email: martina.luzzi@unical.it

Abstract

This study proposes an integrated explanatory–predictive–prescriptive framework for supporting strategic decisions in tourism startups.

Keywords

Abbreviations

Some journals require a list of abbreviations: these normally should be given immediately after the keyswords (if required).

Introduction

The introduction should frame the relevance of innovation and technological resources in the tourism sector, with specific attention to tourism startups. It should emphasize that startups often face resource constraints and strategic uncertainty when deciding where to invest to enhance performance and revenue growth. The section should also highlight the limitations of traditional empirical studies, which mainly explain relationships among variables but rarely translate empirical findings into prescriptive decision-support tools. This motivates the methodological gap addressed by the study: the limited integration of structural equation modeling, configurational analysis, machine learning, and operations research within a unified framework.

While traditional revenue management focuses on pricing and capacity allocation, tourism startups increasingly face strategic decisions regarding how to allocate scarce innovation and technological resources to maximize growth and performance. The proposed framework extends the logic of revenue management from operational pricing decisions to strategic innovation allocation decisions aimed at enhancing revenue growth in tourism startups.

Literature Review

Open Innovation in Tourism Startups

This subsection should discuss the role of open innovation in tourism startups, focusing on knowledge exchange, external collaboration, innovation ecosystems, and the relationship between collaborative innovation practices and firm growth.

Technological Resources and Innovativeness

This subsection should review the role of technological resources as strategic assets for tourism startups. Specific attention should be given to advanced ICT, Industry 5.0 technologies, green technologies, and their contribution to innovation capability and business model development.

Revenue Growth in Tourism Startups

This subsection should focus on revenue growth as a key performance outcome for tourism startups. It should address issues related to scaling, digital transformation, competitiveness, and the ability of startups to convert innovation and technology adoption into measurable economic performance.

Methodological Background

This subsection should introduce the methodological foundations of the study, including covariance-based structural equation modeling, fuzzy-set Qualitative Comparative Analysis, machine learning approaches in management research, and prescriptive analytics based on operations research.

The literature review should build toward the following research gap: Previous studies mainly adopt explanatory approaches, while integrated explanatory–predictive–prescriptive frameworks remain underexplored.

Framework

The proposed framework adopts an integrated explanatory–predictive–prescriptive approach to investigate how open innovation practices, technological resources, and innovativeness influence revenue growth in tourism startups.

As depicted in the graph reported in Figure 1 the framework starts from a theory-driven conceptual model in which Open Innovation Practices, Technological Resources, and Innovativeness are considered key drivers of startup performance, while Revenue Growth represents the main outcome variable. The empirical analysis is conducted on a dataset of 311 tourism startups.

To capture the phenomenon from complementary analytical perspectives, the framework combines three methodological approaches.

First, partial least squares structural equation modeling (PLS-SEM) is employed to validate the conceptual model and estimate the structural relationships among constructs. This step provides an explanatory perspective by identifying the average net effects and testing the hypothesized causal relationships.

Second, fuzzy-set Qualitative Comparative Analysis (fsQCA) is applied to identify alternative configurations of conditions leading to high revenue growth. Unlike SEM, which focuses on linear net effects, fsQCA captures equifinality and causal complexity, showing how different combinations of innovation and technological resources may generate successful outcomes.

Third, XGBoost is adopted as a predictive analytics technique to evaluate the predictive relevance of the variables and uncover nonlinear effects and interaction patterns. Feature importance measures further support the identification of the most influential drivers of revenue growth.

The insights generated by the explanatory, configurational, and predictive analyses are then integrated into a prescriptive decision-support layer based on Operations Research and optimization modeling. In this phase, innovation and technology variables are treated as managerial decision levers, while the optimization model identifies the combinations of strategic investments that maximize expected revenue growth under organizational and budget constraints.

Overall, the framework moves beyond traditional explanatory approaches by integrating causal explanation, configurational reasoning, predictive analytics, and prescriptive decision support into a unified methodological architecture for tourism startup analysis.

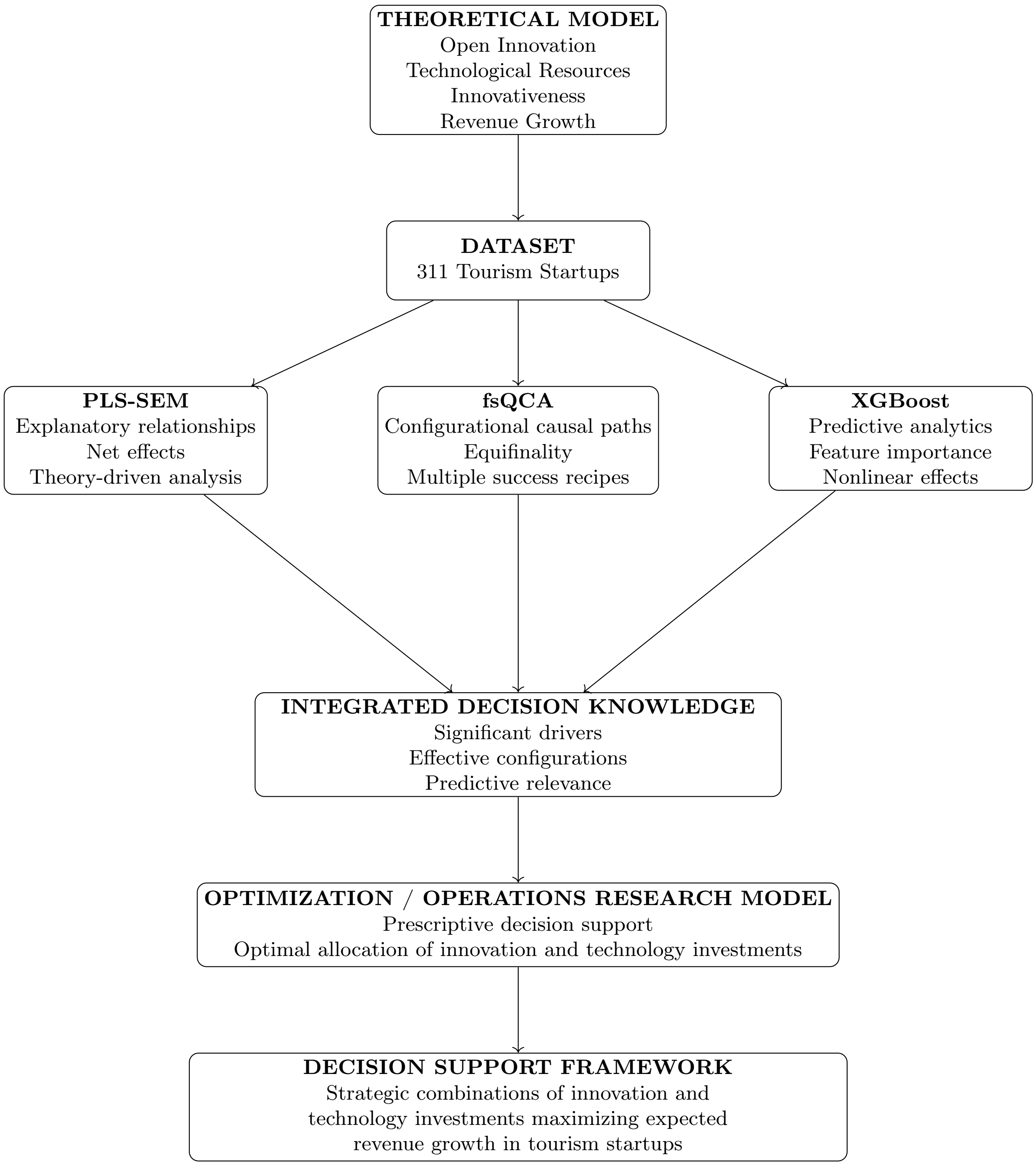

Figure 1: Integrated explanatory–predictive–prescriptive framework

SEM explains average structural effects; fsQCA identifies alternative causal recipes; XGBoost assesses predictive relevance; optimization translates the evidence into prescriptive decision support.

Comparison with traditional Revenue Management:

Table 1: From traditional revenue management to strategic innovation-driven revenue growth

Traditional Revenue Management	Proposed Framework
Allocation of rooms, seats, or capacity	Allocation of innovation and technological resources
Pricing optimization	Innovation investment optimization
Yield management	Strategic growth optimization
Demand–capacity matching	Innovation–performance alignment
Operational decision support	Strategic prescriptive analytics
Revenue maximization from existing assets	Revenue growth through innovation-driven resource allocation

The proposed framework extends the logic of revenue management from short-term operational decisions on pricing, capacity, and demand allocation to strategic decisions on how tourism startups should allocate innovation and technological resources to enhance revenue growth.

Conceptual Model and Hypotheses

Methodology

Dataset and Variables

This subsection should describe the empirical dataset, composed of 311 tourism startups. It should explain the survey-based data collection process and the operationalization of the main variables, including open innovation practices, technological resources, innovativeness, revenue growth, and firm size.

PLS-SEM Analysis

fsQCA Analysis

XGBoost Predictive Analysis

Optimization Model

This subsection should present the prescriptive component of the study. The optimization model should translate innovation and technology-related variables into managerial decision levers. It should define the decision variables, objective function, constraints, and scenario analysis.
The model may aim to maximize expected revenue growth subject to budget, organizational capacity, and firm-specific constraints.

Results

SEM Results

This subsection should present the results of the PLS-SEM analysis, including measurement model validation, structural path coefficients, statistical significance, and model fit indices.

fsQCA Results

This subsection should report the configurations leading to high revenue growth, highlighting alternative causal paths and the role of equifinality.

XGBoost Results

This subsection should present the predictive performance of the XGBoost model, the most relevant predictors, and the interpretation of feature importance and SHAP values.

Optimization Scenarios

This subsection should report the results of the optimization model, including optimal combinations of innovation and technology investments, budget allocation scenarios, and strategic implications for different types of tourism startups.

Discussion

Theoretical Implications

Managerial Implications

Conclusion

Acknowledgements

Please use "The authors thank ..." rather than "The authors would like to thank ...".

Proposte titoli

- From Explanation to Prescription: An Integrated SEM–fsQCA–Machine Learning–Optimization Framework for Revenue Growth in Tourism Startups
- Explaining, Predicting, and Prescribing Growth: A Hybrid Analytics Framework for Tourism Startups
- Open Innovation, Technological Resources, and Revenue Growth in Tourism Startups: An Integrated Explanatory–Predictive–Prescriptive Approach
- Hybrid Analytics for Tourism Startups: Integrating SEM, fsQCA, Machine Learning, and Optimization for Strategic Decision Support
- Beyond Explanation: A Prescriptive Analytics Framework for Strategic Growth in Tourism Startups
- From Innovation Drivers to Strategic Decisions: A Multi-Method Framework for Revenue Growth in Tourism Startups
- Beyond Traditional Revenue Management: A Hybrid Analytics Framework for Strategic Growth Optimization in Tourism Startups
- Strategic Revenue Optimization in Tourism Startups: An Integrated SEM–fsQCA–Machine Learning–OR Framework
- Beyond Traditional Revenue Management: Explaining, Predicting, and Optimizing Growth in Tourism Startups
- From Innovation to Growth: A Hybrid Analytics Framework for Tourism Startups
- From Explanation to Prescription: Strategic Revenue Optimization in Tourism Startups
- Rethinking Revenue Management: Innovation, AI, and Prescriptive Analytics for Tourism Startups
- From Explanation to Prescription: A Hybrid Analytics Framework for Strategic Growth in Tourism Startups